Bcho NAM Player

VST3 / AU Plugin Manual



VST3 / AU Plugin | Version 1.8.5

Complete reference for the 1.8.5 plugin: installation, routing, NORMAL / PLUS, NAM/IR files, controls, all effect parameters, HARMONIZER, presets and automation. The standalone-only DI player is not part of the plugin.

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1. What the plug-in is

An **audio-effect** plug-in - not a MIDI instrument. It carries two complete and independent signal paths (L and R), each with (in NORMAL) BLOCK NAM 1, BLOCK NAM 2, a cabinet IR, an eight-effect rack with free ordering, gate, tone stack, master level, IR blend, IR volume, input and output gain, power, calibration and tuner.

Format	Platforms
VST3	Windows x64, macOS Intel and Apple Silicon, Linux x86_64
AU	macOS only

AAX is not included. Match the package to your DAW's process architecture.

2. Installing

Choosing EmbeddedBrowser or ExternalBrowser

Version 1.8.5 provides two variants for both **Standalone** and **VST3** on Windows x64:

ZIP variant	TONE3000 sign-in and catalogue	Browser requirement
EmbeddedBrowser	Opens sign-in pages and SEARCH FULL CATALOGUE inside the application	Microsoft WebView2
ExternalBrowser	Opens those pages in your default system browser	Does not require WebView2

The TONE3000 library remains available through the TONE3000 button in both variants. The audio engine, controls, NAM models, IRs and presets are the same; only the browser used for sign-in and catalogue selection changes. Local files work offline in both.

Choose a **Standalone** ZIP to run the application, or **VST3** to use it in your DAW. Both VST3 variants share the same plug-in name and identity: install only one at a time. To switch, close the DAW, replace the entire Bcho NAM Player.vst3 folder and rescan plug-ins. Do not combine files from the two packages. You can keep both standalone variants in separate folders. Keep your models, IRs and presets when replacing a variant.

The cover image is a standalone screenshot; the plug-in manual describes the controls available inside a DAW. Embedded browser examples refer to EmbeddedBrowser; in ExternalBrowser, complete those steps in your system browser.

Close the DAW first, and copy the **entire bundle**, not just the binary inside it.

System	Destination
Windows	C:\Program Files\Common Files\VST3 (administrator permission may be required)
macOS VST3	~/Library/Audio/Plug-Ins/VST3
macOS AU	~/Library/Audio/Plug-Ins/Components
Linux	~/.vst3

Start the DAW and rescan plug-ins. The effect appears as **Bcho NAM Player**.

macOS builds are ad-hoc signed, not notarized. If the host blocks the plug-in, review Privacy & Security and your host's own validation - do not disable system protection globally.

3. Your first track

1. Insert the plug-in as an **effect** on a mono or stereo audio track.
2. Select your guitar input in the DAW and enable monitoring. Avoid monitoring the same signal twice, through the interface and the DAW.
3. A new instance starts in **MONO** with every block disabled, so your dry signal passes through until you load a capture. POWER can mute the path.
4. Load a capture with **BROWSE LOCAL** or **TONE3000**, then a cabinet with **BROWSE IR**.
5. Set levels with INPUT GAIN, MASTER VOL and OUTPUT GAIN while watching the meters.

Driver, sample rate and buffer size belong to the DAW. The plug-in has no audio-device window.

4. Map of the editor



The plug-in editor in MONO

The layout is the standalone's, minus the parts a host owns:

- **Rack strip**: preset buttons, NORMAL / PLUS, the processing blocks, the tuner and its display and CHROMATIC indicator, the MONO / STEREO switch and the settings gear.
- **Amplifier head**: Input VU, INPUT GAIN, INPUT CALI; the seven main knobs with the NAM and IR selectors above them; the NAM and IR browsers with the IR VOL fader; Output VU, OUTPUT GAIN and POWER.
- **Cabinet**: the logo and two cones driven by the real output level.

The editor is fully resizable and keeps its 1537 x 1023 proportions.

5. MONO, STEREO and the channel layout

The plug-in always presents a **stereo output**, even on a mono track, so both chains always have somewhere to go.

Mode	Behaviour
MONO	Only the left path runs. Its result is copied to both plug-in outputs
STEREO	Left path processes input channel 1 and feeds output 1; right path processes input channel 2 and feeds output 2

On a **mono track in STEREO**, both chains receive the same input signal, so one guitar drives two independent rigs.
On a **stereo track in STEREO**, the two input channels are processed separately.



The plug-in in STEREO: two rack rows, four NAM selectors, two IR selectors

The standalone's DUAL MONO / SPLIT L/R switch is not present: in a DAW you decide that with the track's own routing.

PAN (the bar between the two rack rows and the tuner, STEREO only): a horizontal fader that balances the level of the two paths.

Position	Result
Centre (CENTER, centre notch, green mark)	L and R play at their full level, exactly as without PAN
Towards L	The right path fades out progressively; at the left stop (L 100 %) only L is heard
Towards R	The left path fades out progressively; at the right stop (R 100 %) only R is heard

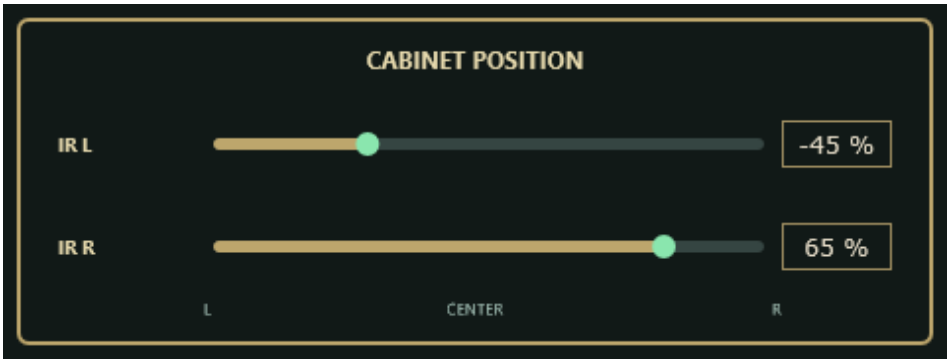
The L and R letters at the ends are lamps: the one on the side being faded dims, and the rail lights up from the centre towards the favoured side. Neither side ever gets louder than it is at the centre. Double-click returns the fader to the centre; the mouse wheel moves it in fine steps.

In the plug-in PAN is an automatable parameter (Pan) and is saved with the project.

OUT is a shared post-chain output trim (-60 to +12 dB, 0 dB by default). It changes both processed channels together without changing their relative balance. Double-click the strip to return to 0 dB. In the standalone it affects MAIN and WET, not the PRE and DI reference taps. In the plug-in it acts on the final stereo output.

Cabinet position

CAB POS places the two complete processed cabinet paths in the final stereo field. Its **IR L** slider moves the left cabinet path and **IR R** moves the right one. Initially they are hard left and hard right. Move IR L to CENTER to hear that cabinet in both speakers while IR R remains on the right; center both to hear both cabinets from both speakers. These sliders do not select or blend IR files. In PLUS, two IRs in a lane run in series: CAB POS moves the lane's combined result, not each IR separately. **PAN** subsequently balances the L and R paths; **OUT** changes the final level of both together; **OUTPUT GAIN** belongs to the selected path. Position moves are smoothed to avoid clicks. The two positions and OUT are automatable and saved with the DAW project.



Cabinet position controls for the left and right processed paths

The L / R tabs beside the racks choose which path the shared controls edit; the NAM 1 L, NAM 2 L, NAM 1 R, NAM 2 R and IR L, IR R selectors work exactly as in the standalone - see Which control follows what.

6. Loading captures and cabinets

Identical to the standalone (chapter 8 and chapter 9):

- Choose the target block with the NAM selectors, then **BROWSE LOCAL**, the list, **TONE3000**, a double-click on the block, or drag and drop.
- **BROWSE IR** loads a .wav response; clicking the selected row again clears it and bypasses the IR block.
- Loading a valid file switches its block on. IR gain is never normalized.
- NAM A1, A2 Standard and A2 Nano are detected automatically.
- Hovering a NAM block or list row shows the capture information card.

Demonstration captures ship in the Models folder beside the package but are never loaded automatically; find them with BROWSE LOCAL.

> The DAW project stores **references** to your .nam and .wav files, not the files themselves, including PLUS AIFF/FLAC IR files. Keep them in place, or save .bnpp presets, which embed them.

7. Controls

Operating a knob: drag vertically or horizontally; the exact value is printed underneath. **Double-click restores the default** - 12 o'clock for everything except GATE, which returns to fully left (OFF).

Control	Range	Default	Notes
INPUT GAIN	-12 … +12 dB	0.0 dB	Level into the NAM chain; Input Cali is added on top

Control	Range	Default	Notes
GATE	OFF … -80 … 0 dB threshold	OFF	At the far left it is a true bypass. Attack 1.5 ms, hold 35 ms, release 90 ms, 3 dB hysteresis
BASS	±12 dB @ 70 Hz	0.0 dB	Peaking, Q 0.72
MID	±12 dB @ 750 Hz	0.0 dB	Peaking, Q 0.72
TREBLE	±12 dB @ 4 kHz	0.0 dB	Peaking, Q 0.72
PRESENCE	±12 dB @ 6 kHz	0.0 dB	Peaking, Q 0.72
MASTER VOL	-12 … +12 dB	0.0 dB	After the chain, before Output Gain
IR BLEND	0 … 100 %	50 %	Dry against cabinet
IR VOL	-24 … 0 dB	-12 dB	Cabinet branch only
OUTPUT GAIN	MUTE, then -60 … +12 dB	0.0 dB	Final level of the selected path; centre is unity
POWER	On / off	On	Mutes the processed path; the switch glows red while on
INPUT CALI	On / off	Off	See Input Cali in chapter 10

At the far-left stop, **OUTPUT GAIN** displays **MUTE** and the selected path becomes completely silent after a 10 ms ramp. Above the stop, the lower half rises from -60 dB to 0 dB at the centre; the upper half reaches +12 dB. L and R can be muted independently or together with their link. If CAB POS places the other, unmuted path in the same physical speaker, that speaker may still sound: OUTPUT GAIN mutes a path, not a speaker. Link L/R and turn the knob to MUTE to silence both paths.

The tone stack follows the selected NAM block. With a **NAM 1** selector active, BASS / MID / TREBLE / PRESENCE drive BLOCK NAM 1's own four filters; with a **NAM 2** selector active they drive the main tone stack. Each set keeps its own values, and switching selectors recalls them. In the plug-in and in STEREO the plate prints which one you are editing.

In STEREO, INPUT GAIN and OUTPUT GAIN show (L) or (R) beside their labels. Each path-dependent front-panel control has its own **L-chain-R** switch, except TUNER. Select L or R, set its value, then light that parameter's link: the current value is copied to the other path, and later editor changes to that parameter update both. Turn the link off to edit paths independently; their current values remain in place. Each parameter has a separate link. The link beside POWER makes its red switch turn both paths on or off together; it does not link audio, gain or tone. INPUT CALI has its own link beside its indicator. Tone links follow the selected NAM block on each path, including the viewed NAM slot in PLUS. TUNER listens only to the selected path and has no link. Link choices are saved with the DAW project; rack effect links remain separate.

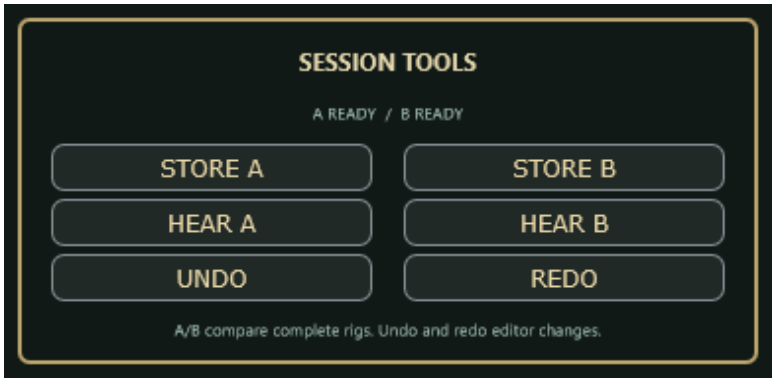
Control	STEREO scope
INPUT GAIN, GATE, MASTER VOL, OUTPUT GAIN, POWER, INPUT CALI	Selected L or R path; each has its own optional link
BASS, MID, TREBLE, PRESENCE	Selected NAM slot on the selected path; each has its own optional link
IR BLEND, IR VOL	Selected IR L or IR R cabinet; each has its own optional link
TUNER	Selected path only; no link and no mixing of two notes
PAN, OUT	Both processed paths together; independent of the L/R path tab
CAB POS IR L / IR R	Independent placement of each complete cabinet path

The L/R links are editor gestures. The DAW still exposes its own L_ and R_ parameters for host automation; automating one of those parameters addresses that side directly.

RESET NAM, beside NAM MODELS, offers **NAM ONLY** (selected NAM tone and the path's MASTER VOL / OUTPUT GAIN) or **ENTIRE PATH** (all front-panel controls on the selected path, plus its NAM tones and slot levels in PLUS). The confirmation spells out that loaded NAMs, IRs and effect order are kept. Lit links also carry reset values to the other path. An IR slot cannot be reset as a NAM, so the button is disabled while an IR slot is selected in PLUS.

Session tools: A/B and history

The gold-edged **SESSION TOOLS** button beside IR FILES opens a closable window without covering the rack. **STORE A/B** capture two complete plug-in rigs; **HEAR A/B** recall their NAMs, IRs, effects, levels, routing, stereo placement and NORMAL / PLUS chains. No automatic loudness matching is applied. **UNDO** and **REDO** move through up to 40 recent rig changes; continuous knob drags are grouped after they settle. A/B recalls are undoable. These temporary working copies belong to the open editor and are not written into the DAW project or .bnpp files. This history is separate from the DAW's own undo history and its parameter automation. File changes use the existing output fade.



Session tools in the plug-in

8. The rack and the eight effects

Identical to the standalone (chapter 7 and chapter 11): single click enables or bypasses, double-click opens the editor or the file chooser, drag reorders around the BLOCK NAM 2 and IR anchors, and the eye chooses which block the shared controls describe.



The effect editor

Compressor, octaver, pitch shifter, chorus, flanger, phaser, delay and reverb, each with three algorithms and six real-unit parameters. The complete parameter tables and HARMONIZER instructions are in Appendix A below.

NORMAL / PLUS: NAM and IR chains

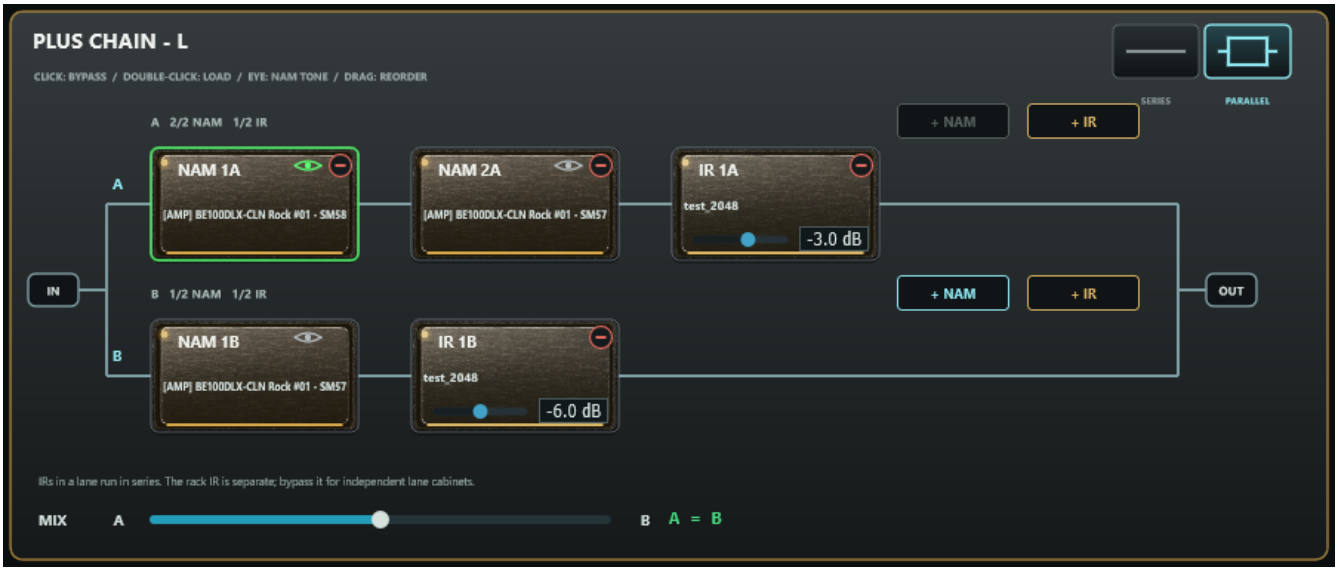
The lever switch on the left end of the rack chooses between two ways of using NAM captures:

Position	What the rack shows
NORMAL (lever down)	BLOCK NAM 1 and BLOCK NAM 2, exactly as described above
PLUS (lever up)	One large **NAM / IR** block, twice as wide, at BLOCK NAM 1's position in the chain



PLUS: one double-width NAM / IR block per path, with its chain topology under the name

The NAM / IR block of PLUS runs a **chain** of NAM and cabinet IR blocks. It switches on and off with a click and has the eye like any other NAM block; the line under its name shows the chain, for example SERIES 2 or PARALLEL 2 | 1. **Double-click it** to open the PLUS CHAIN window:



The PLUS CHAIN window in parallel: lane A above, lane B below, the + buttons and the A / B MIX

Element	Use
Single-line / parallel-lines switch	**SERIES**: one lane from IN to OUT. **PARALLEL**: two lanes, A above and B below, fed by the same input and mixed back together
NAM 1A / 2A and IR 1A / 2A (likewise B)	Up to two NAMs and two IRs per lane. Each lane retains one NAM placeholder. Blocks run in the displayed order
Click on a block	On / off (an empty block opens the load menu instead)
Double-click or right-click	NAM: load a local capture or use TONE3000. IR: load a local WAV, AIFF or FLAC. Remove block is available when allowed
Drop a file	Drop .nam on a NAM card or + NAM; drop WAV, AIFF or FLAC on an IR card or + IR
Eye (NAM cards only)	Select the NAM block edited by BASS, MID, TREBLE, PRESENCE and loaded from the NAM MODELS list or TONE3000
- (top right corner)	Remove the block, after confirmation. The last NAM of a lane cannot be removed; all IRs can be removed
+ NAM / + IR	Add the chosen type; each button becomes unavailable at its two-block limit. New NAMs are inserted before IRs
IR level	-24 to +12 dB, initially 0 dB; double-click to reset. WAV, AIFF and FLAC, mono or stereo averaged to mono, up to 8192 source samples, resampled without normalization
Drag a block	Reorder it, also into the other lane (if its two-NAM / two-IR limits allow it)
MIX A - B (PARALLEL only)	Linear crossfade: centre gives 50% of each lane; the ends give 100% of A or B. Unlike PAN, centre is not full level on both lanes

Each NAM block keeps its own capture, on/off switch and tone controls, just like BLOCK NAM 1 and BLOCK NAM 2. In PLUS the NAM selectors above the tone plate and over the list become one per path, captioned with the block being edited (NAM 2A), and the list shows the NAM models of that block.

The first time PLUS is switched on, loaded NORMAL NAM captures are copied into lane A in rack order, with their bypass and tone settings. An empty NAM placeholder remains if neither capture is loaded. The general rack IR is not copied into a lane. NORMAL and PLUS keep separate settings: going back to NORMAL finds BLOCK NAM 1 and BLOCK NAM 2 exactly as they were. The switch, the chains and their captures are saved with the session, in presets (the .bnpp bundle embeds every chain capture and IR) and, in the plug-in, with the DAW project. Every change of the chain is applied behind a short fade, with a short transition to suppress clicks. With INPUT CALI on, PLUS calibrates to the first active NAM of lane A; standalone DI playback bypasses interface calibration.

IRs within a lane run in series. To mix two cabinets, put one in each parallel lane and use A/B MIX. The rack IR remains a separate shared block: bypass it when using independent lane cabinets to avoid filtering the signal twice. IR files load in the background, and their level, order and bypass state are saved with the chain. Legacy chains with

more than two NAMs per lane restore the first two and show a notice; keep the original preset if you need its older configuration.

The PITCH block includes a scale-aware **HARMONIZER** (thirds and fifths that follow the key, octaves, AUTO key detection): see HARMONIZER in Appendix A.

9. Linking effects between L and R

In STEREO the chain icon links an effect of L with the same effect of R, so one edit changes both. Green = linked, grey = not. When the two blocks share a column the icon sits on the seam between the rows; when they do not, it moves to the lower-right corner of both blocks.



Linked and unlinked pairs in the plug-in's two rack rows

Linked blocks share the algorithm and the six parameters; each keeps its own on/off switch and its own position. Linking two blocks that differ asks which side's settings to keep. BLOCK NAM 1, BLOCK NAM 2 and IR cannot be linked. **Links are saved in the DAW project.** Full description: standalone chapter 13.

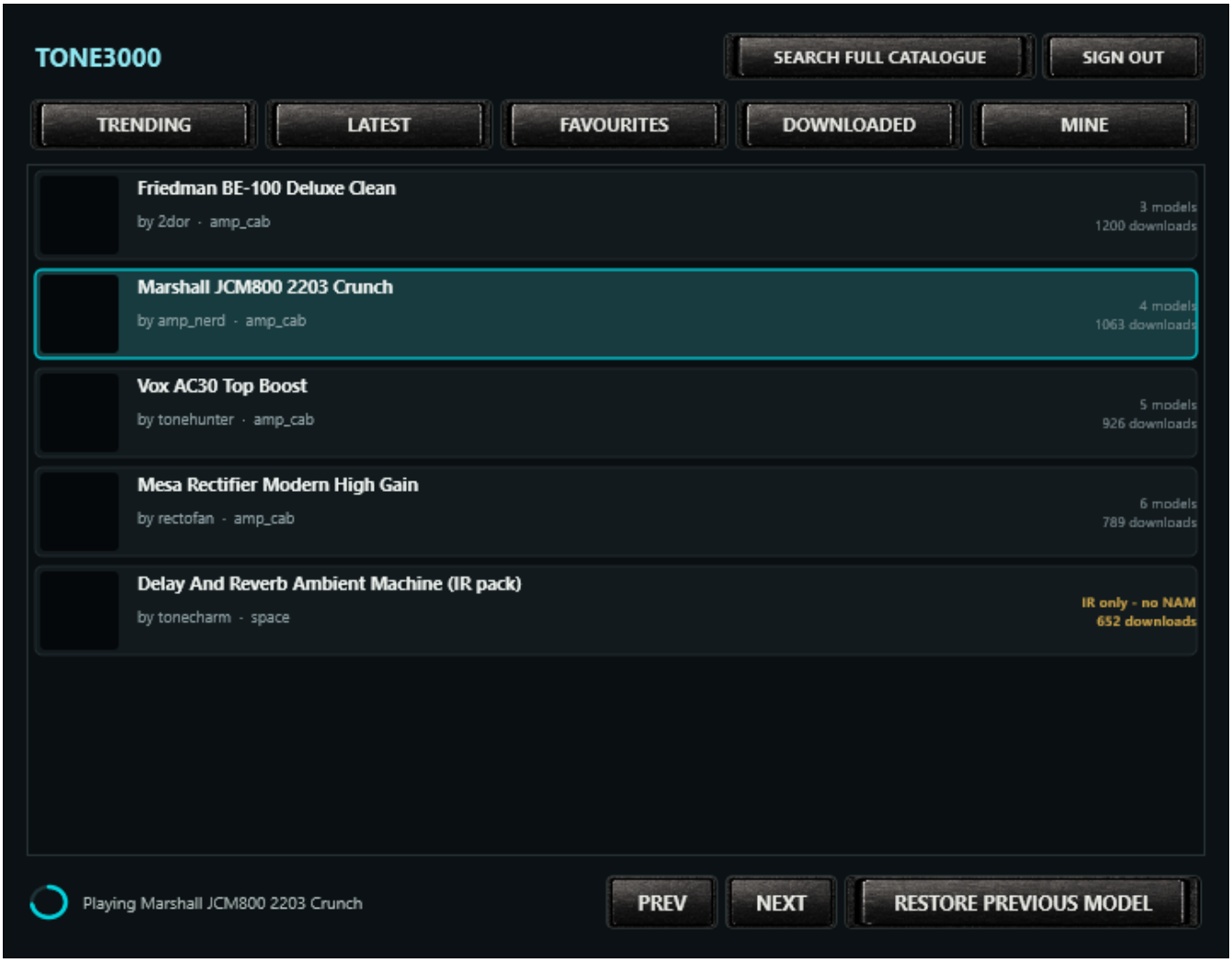
10. Tuner and Input Cali

TUNER listens to the untouched input before the gate and NAM blocks, covers roughly 65-700 Hz and reads centred within ±5 cents. **CHROMATIC** reports the detected note regardless of guitar tuning; it does not assign a string or target note. Compare its reading with the note you intend for the string. Play one isolated string at a useful level. In STEREO it monitors the selected L or R path, never both notes mixed; each path keeps its own tuner on/off state.

INPUT CALI uses the engine's reference and the capture's input_level_dbu metadata (assuming +12 dBu when a capture has none), clamped to ±24 dB. The plug-in does **not** expose the standalone's interface-reference slider, because the interface belongs to the DAW: for a specific hardware calibration, set the gain outside the plug-in or use the standalone.

11. TONE3000 inside the plug-in

The **TONE3000** button opens the same library window as the standalone.



The TONE3000 library

- Connect the account once; the encrypted refresh token is stored per user, so subsequent connections can reuse it while authorization remains valid. Reconnect if it expires or is revoked.
- TRENDING, LATEST, FAVOURITES, DOWNLOADED and MINE.
- **Selecting a row loads that capture into the current NAM block immediately**; the rest of the tone downloads in the background, and picking another row cancels it. A tone already in your downloads loads instantly.
- Rows marked **IR only - no NAM** hold only impulse responses and are not auditioned.
- **RESTORE PREVIOUS MODEL** puts back what the block held before the window was opened; closing the window keeps the last audition.
- **SEARCH FULL CATALOGUE** opens TONE3000's picker inside the plug-in (EmbeddedBrowser) or in the system browser (ExternalBrowser). Choose a tone to start its download; the embedded picker closes automatically.

On Windows the embedded browser uses WebView2. Inside a DAW the plug-in stores its WebView2 data in a per-user folder, so a host installed in Program Files never blocks it. If WebView2 is missing, the plug-in offers an official Microsoft download or your external browser and installs nothing automatically. Full detail: standalone chapter 16.

12. Projects, presets and automation

The DAW project stores everything about the instance: both paths, every control, the file references, rack order, algorithms, parameters, bypass states, the L/R links, and MONO/STEREO.

Reopening a project re-loads the saved NAM, pedal NAM and IR of **both** paths into the engine with their saved on/off states - whatever order the host uses to restore the state and prepare audio, and with the editor closed. Project restore, capture changes and audio re-initialisation (a sample-rate or buffer-size change) all fade in from silence, so nothing clicks.

.bnpp **presets.** SAVE PRESET writes a portable archive for the **selected path** with its NORMAL and PLUS NAM/IR files and settings embedded and checked with SHA-256; LOAD PRESET replaces that path. Share both paths as two presets; use the project to recall the whole instance. A two-path standalone preset loads its left-path representation into the selected plug-in path.

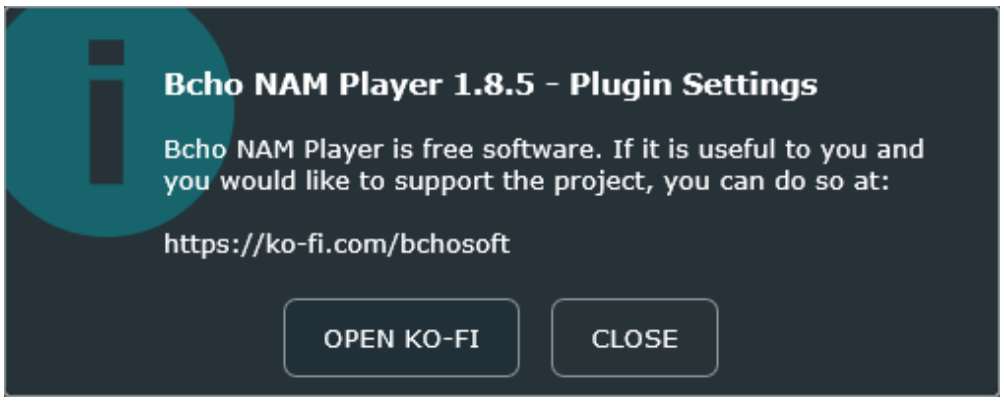
Automation. The host can automate the published parameters, per path:

Parameter	Per path
Input Gain, Gate, Bass, Mid, Treble, Presence, Master Vol	L_ and R_
NAM 1 Bass, NAM 1 Mid, NAM 1 Treble, NAM 1 Presence	L_ and R_
IR Blend, IR Cabinet Volume, Output Gain	L_ and R_
Power, Input Cali, Tuner	L_ and R_
Stereo (MONO / STEREO)	one, global
Pan (only acts in STEREO)	one, global
Global Output (OUT)	one, global
Cabinet Position L / R (CAB POS, only acts in STEREO)	one per processed path

Rack order, effect algorithms and effect parameters are part of the saved state, not host automation. PLUS mode, topology, per-block files, tone, levels and bypass are saved in state, without adding host automation parameters. DI playback adds no controls, parameters or state to the plugin. Loading a file is a load, not a continuous parameter.

13. Settings and support

The gear opens a short window with the version and a link to support the project.



Plug-in settings

14. Differences from the standalone

The standalone has	The plug-in
Audio Setup: driver, interface, channels, sample rate, buffer	The DAW owns all of it
Physical MAIN / PRE / DI / WET output routing	Route inside the DAW
DUAL MONO / SPLIT L/R input switch	Decided by the track's routing
Interface input reference slider for Input Cali	Not exposed; calibrate outside or use the standalone
Fifteen selectable skins, including the green, blue and graphite finishes	Always Astra / Obsidian
Its own saved session between launches	State comes from the host project
Update checking	Not included
IN / DI player, CONFIG DI, transport and DI file state	Not included. Play and route DI audio tracks in the DAW
REC take review, DI / PROCESSED WAV / MP3 export and metronomes	Not included. Record, count in and monitor the metronome in the DAW

Everything else - engine, rack, effects, links, tuner, browsers, TONE3000, information card, .bnpp presets - is the same code.

15. Troubleshooting

Symptom	What to check
Not listed after scanning	Folder, format and architecture match the host; then rescan
No audio	Track input, monitoring, POWER, and the DAW's own routing
Dry sound only	A capture is loaded and its block is lit; a fully bypassed chain passes dry audio
One side silent in STEREO	The DAW must actually feed both channels; on a mono track both chains get the same input
Missing files after moving a project	Restore the .nam / .wav locations, or load your .bnpp presets
Dropouts	Raise the buffer and lower CPU load; PLUS can run eight NAM blocks across two paths, in addition to IRs and effects
Empty TONE3000 window	Account, network, or WebView2 on Windows; or use the external browser
macOS host rejects it	Review host validation and system permissions; ad-hoc signing is not notarization

16. Technical specifications

Item	Value
Plug-in type	Audio effect (no MIDI)
Bus layout	Mono or stereo input, **always stereo output**
Signal paths	2 independent (L / R)
NORMAL	2 NAM and 1 rack IR per path; 8 effects per path
PLUS	2 NAM + 2 IR per lane, up to 2 lanes per path; general rack IR remains separate
Internal version	1.8.5 (numeric version 0x10805); VST3 and AU metadata derive from the same project version
Sample rates / buffers	Whatever the host provides; oversized host buffers are split internally
NAM architectures	A1, A2 Standard, A2 Nano - automatic
Start-up / re-prepare fade	60 ms from silence
NAM change fade	15 ms out, silence until the new capture is installed, 20 ms in
IR change fade	12 ms crossfade
Editor	1537 x 1023 native, resizable, proportional

17. Integrity and licences

Check ZIP hashes against SHA256SUMS.txt and read THIRD_PARTY_NOTICES.md (JUCE and Neural Amp Modeler Core). Respect the licences of the captures and impulse responses you use or share inside presets. These manuals do not imply certification in every DAW; each host and system combination deserves a practical check.

Appendix A. Complete effects reference

The eight effects offer **three algorithms** and **six main parameters** each. PITCH also includes the HARMONIZER controls described below. Double-click a block to open its editor; every knob shows a real unit, and double-clicking a knob restores its default.



The effect editor, here for DELAY

The **ACTIVE** switch at the top right of the editor is the same bypass as clicking the block. **TYPE** chooses the algorithm. Parameter, algorithm and bypass changes are smoothed, delay-family reads are interpolated and feedback is bounded, so nothing clicks or runs away.

COMP - compressor

Algorithms: **Studio VCA**, **Optical**, **FET Punch**.

Parameter	Range	Default
Threshold	-55 … -2 dB	-31.2 dB
Ratio	1 … 20 :1	7.7:1
Attack	1 … 100 ms	15.9 ms
Release	20 … 600 ms	223 ms
Makeup	-12 … +12 dB	0.0 dB
Mix	0 … 100 %	100 %

DELAY

Algorithms: **Digital Studio**, **Tape Echo**, **Analog BBD**.

Parameter	Range	Default
Time	20 … 1200 ms	398 ms
Feedback	0 … 92 %	32 %
Mix	0 … 100 %	28 %
Tone	0 … 100 %	65 %
Mod	0 … 100 %	8 %
Level	-12 … +12 dB	0.0 dB

CHOR - chorus

Algorithms: **Studio**, **Ensemble**, **Tri-Chorus**.

Parameter	Range	Default
Rate	0.05 … 5 Hz	1.24 Hz
Depth	0.5 … 20 ms	10.3 ms
Mix	0 … 100 %	35 %
Delay	4 … 30 ms	11.8 ms
Feedback	-65 … +65 %	0 %
Level	-12 … +12 dB	0.0 dB

FLANG - flanger

Algorithms: **Analog**, **Through-Zero**, **Jet**.

Parameter	Range	Default
Rate	0.03 … 2.5 Hz	0.57 Hz
Depth	0.1 … 9 ms	5.0 ms
Mix	0 … 100 %	35 %
Feedback	-85 … +85 %	20 %
Manual	0.2 … 5 ms	1.4 ms
Level	-12 … +12 dB	0.0 dB

PHASE - phaser

Algorithms: **4 Stage**, **8 Stage**, **12 Stage**.

Parameter	Range	Default
Rate	0.03 … 4 Hz	0.82 Hz
Depth	0 … 100 %	70 %
Mix	0 … 100 %	40 %
Feedback	-75 … +75 %	12 %

Parameter	Range	Default
Centre	180 … 2200 Hz	887 Hz
Level	-12 … +12 dB	0.0 dB

REVERB

Algorithms: **Studio Room**, **Plate**, **Concert Hall**.

Parameter	Range	Default
Size	0 … 100 %	55 %
Damping	0 … 100 %	50 %
Mix	0 … 100 %	25 %
Width	0 … 100 %	80 %
Freeze	OFF / ON	OFF
Level	-12 … +12 dB	0.0 dB

OCT - octaver

Algorithms: **Poly Clean**, **Classic Mono**, **Organ**.

Parameter	Range	Default
Oct Down	0 … 100 %	45 %
Oct Up	0 … 100 %	0 %
Dry	0 … 100 %	80 %
Tone	0 … 100 %	50 %
Tracking	0 … 100 %	65 %
Level	-12 … +12 dB	0.0 dB

PITCH - pitch shifter

Algorithms: **Studio**, **Low Latency**, **Vintage**.

Parameter	Range	Default
Semitones	-12 … +12 st	0.0 st
Mix	0 … 100 %	100 %
Window	20 … 120 ms	65 ms
Feedback	0 … 55 %	0 %
Fine	-100 … +100 ct	0 ct
Level	-12 … +12 dB	0.0 dB

HARMONIZER (scale-aware harmony)



The PITCH window with the HARMONIZER strip: switch, INTERVAL, KEY, SCALE, TUNING and the live display

The strip at the bottom of the PITCH window turns the block into an intelligent harmonizer. With **HARMONIZER** on, every note you play gets a second voice a scale step away, using the selected or detected scale:

INTERVAL	Harmony voice
OCTAVE UP / OCTAVE DOWN	Always an octave; needs no key, sounds from the first note and also works on chords
THIRD UP / THIRD DOWN	The third of the scale: **major or minor depending on the note**. In C major, C gets E (major third) and D gets F (minor third); in C minor, C gets Eb
FIFTH UP / FIFTH DOWN	The fifth of the scale: perfect, or diminished on the seventh degree (B -> F in C major)

KEY and SCALE. With KEY on **AUTO** the key is learned from the notes you play:

- It works on the seven notes in use, which is what decides the harmony. A key is told apart from its parallel (C major / C minor) by the notes actually played (E or Eb, A or Ab, B or Bb); a key, its relative and its modes (C major, A minor, D dorian...) share their notes and therefore their harmony.
- The tonic and the mode are named from where the playing dwells and, above all, where phrases come to rest: the display can read A MINOR, D DORIAN, G MIXOLYDIAN... Pentatonic playing is read as the natural minor / major until other notes say otherwise. **Harmonic minor** is recognised when the raised seventh is used throughout (G# and never G in A minor): the dominant then gets its major third.
- Two memories run side by side: a long one keeps the key steady and a short one follows a real key change within a few seconds. The detector resists short passing notes; fix KEY manually if automatic detection does not match your phrase.
- Until enough notes have been heard the display shows **LISTENING...** and thirds and fifths stay silent, until detection has enough confidence. The bar under the key shows how sure the detection is.

Choose a tonic in **KEY** to fix the key yourself; **SCALE** then offers major, minor, the five other modes, harmonic minor and melodic minor.

TUNING. PURE tunes each interval to the played note with simple ratios (5:4 and 6:5 thirds, 3:2 fifths): the two voices lock together without beating, which is what keeps a harmony clean through an overdriven amp. **TEMPERED** uses the piano's equal semitones, to match keyboards exactly.

The voice. The harmony is made by a shifter of its own whose splices are synchronised to the period of the note being played, so it sounds like a second guitar rather than an effect. It reads the guitar a few milliseconds late - the natural delay of a second player - and uses that time to know each new note before it sounds: a note never comes out with the interval of the previous one. Notes are recognised from low E (and drop tunings) up to the 24th fret, within a few cents, in about 25 ms (45 ms on the lowest strings). Bends and vibrato are followed continuously: in a bend from C to D in C major the harmony slides from E to F. Notes outside the key (a blue note, the G# of A minor) borrow the scale degree that gives a real third or fifth: G# gets B, Bb in C major gets D.

Knobs in HARMONIZER mode: **MIX** balances the harmony against the note you play, which always stays (100 % = both at the same level, 0 % = no harmony); **FINE** detunes the harmony slightly for a wider sound; **LEVEL** works as usual. SEMITONES, WINDOW and FEEDBACK rest. TYPE chooses the character of the voice: **Studio**, **Low Latency** (a shorter delay; very fast phrases may lose a little accuracy on low notes) and **Vintage** (darker). The rack block reads **HARMONY** while the harmonizer is on.

Play single notes: thirds and fifths follow melodies, riffs and solos. For thirds and fifths, use single-note playing. Polyphonic or unclear input can make pitch detection unreliable; the detector may suppress the harmony. Detection uses the clean guitar signal, before gain, NAM and effects. The HARMONIZER settings are saved with the PITCH block and follow an L / R link.

Latency: the original guitar remains on the immediate dry path. The HARMONIZER adds a separate voice whose pitch analysis and splice window cause a short, variable delay; **Low Latency** shortens that voice delay. The plug-in reports **0 samples of fixed whole-plug-in latency** to the DAW. Delaying the entire plug-in to compensate the variable harmony voice would also misalign the dry guitar and the other effects.